You can clearly see the localhost port corresponding to each exposed container machine port. Use the command *bootloose ssh root@<HOSTNAME>* to get into any of the created container machines and run any normal GNU/Linux command there.

You can also run any ssh (or scp) command using the cluster-key private key created in the current directory to transfer and execute commands remotely, to turn your container machines into full-fledged servers, such as the command *bootloose ssh root@kafka1 true && echo ' PING kafka1 response :PONG'*. Finally, you can clean up the created cluster with the command *bootloose delete*.

You should now be able to easily create and use your container machines.

Cloud VMs using Multipass

We have explored how to develop cloud-like virtual machine stacks on our laptops with fat container machines. This didn't require a lot of computing resources and any hardware-based virtualisation support. But security is a downside of containers as these provide thinner isolation in comparison to the strong isolation offered by virtual machines. So is a lightweight and fast solution to create cloud VM stacks on our laptops possible? We also want the tooling that works out-of-the-box without a lot of dependencies. The answer is yes; you can have your cake and eat it too!

The tooling presented in the next section is for newer machines where the virtualisation capability is provided by the processor. Fat containers remain an option in case you don't have such hardware available. I tested the snippets shown or referenced in this article on my 4 core/8GB COREi7 laptop running Ubuntu 18.04 LTS. But I have also used the tool on Mac OS previously in my professional setting.

Multipass usage

Multipass is a tool to build Ubuntu virtual machines that can be configured using cloud-init like public cloud VMs. It's a lightweight virtual machine manager that is cross-platform and uses the standard underlying hypervisors on the respective platform to keep the overhead minimal. So it's possible to build a public cloud-like VM environment anywhere on demand using this tool.

Find an appropriate Ubuntu cloud image to instantiate your first VM. Execute the command *multipass find* to dump all the available and supported images, as shown in Figure 4. You can append the option *--unsupported* in case you are experimenting with old unsupported versions of Ubuntu. You can also filter the images listing by appending the remotes *release:* or *daily:* to the command. And you can get familiar with the default images by including *default* with or without remotes.

Now, we'll look at building local cloud VMs and manipulating them. The command *multipass launch* will bring up a local Ubuntu VM using the default image. Append the desired image/alias to the command to instantiate the VM with that particular image. You need to append the command with *-n <NAME>* to avoid naming your VM instance with an auto generated random name. You also need to append with

```
core                        core16            20200818         Ubuntu Core 16
core18                                        20211124         Ubuntu Core 18
core20                                        20230119         Ubuntu Core 20
core22                                        20230717         Ubuntu Core 22
core24                                        20240603         Ubuntu Core 24
20.04                       focal             20240821         Ubuntu 20.04 LTS
22.04                       jammy             20241002         Ubuntu 22.04 LTS
24.04                       noble,lts         20241004         Ubuntu 24.04 LTS
daily:24.10                 oracular,devel    20241007         Ubuntu 24.10
appliance:adguard-home                        20200812         Ubuntu AdGuard Home Appliance
appliance:mosquitto                           20200812         Ubuntu Mosquitto Appliance
appliance:nextcloud                           20200812         Ubuntu Nextcloud Appliance
appliance:openhab                             20200812         Ubuntu openHAB Home Appliance
appliance:plexmediaserver                     20200812         Ubuntu Plex Media Server Appliance

Blueprint                   Aliases           Version          Description
anbox-cloud-appliance                         latest           Anbox Cloud Appliance
charm-dev                                     latest           A development and testing environment for charmers
docker                                        0.4              A Docker environment with Portainer and related tools
jellyfin                                      latest           Jellyfin is a Free Software Media System that puts you in control of managing
and streaming your media.
minikube                                      latest           minikube is local Kubernetes
ros-noetic                                    0.1              A development and testing environment for ROS Noetic.
ros2-humble                                   0.1              A development and testing environment for ROS 2 Humble.

matrix@ankur-Inspiron-5593:~/Development/Git/Works/devops/CloudInABox/ContainerMachines/scripts$ multipass find default
Image                       Aliases           Version          Description
default                                       20241004         Ubuntu 24.04 LTS
daily:default                                 20241004         Ubuntu 24.04 LTS

matrix@ankur-Inspiron-5593:~/Development/Git/Works/devops/CloudInABox/ContainerMachines/scripts$ multipass find release:default
Image                       Aliases           Version          Description
release:default                               20241004         Ubuntu 24.04 LTS

matrix@ankur-Inspiron-5593:~/Development/Git/Works/devops/CloudInABox/ContainerMachines/scripts$ multipass find daily:default
Image                       Aliases           Version          Description
daily:default                                 20241004         Ubuntu 24.04 LTS
```

Figure 4: Multipass images listing